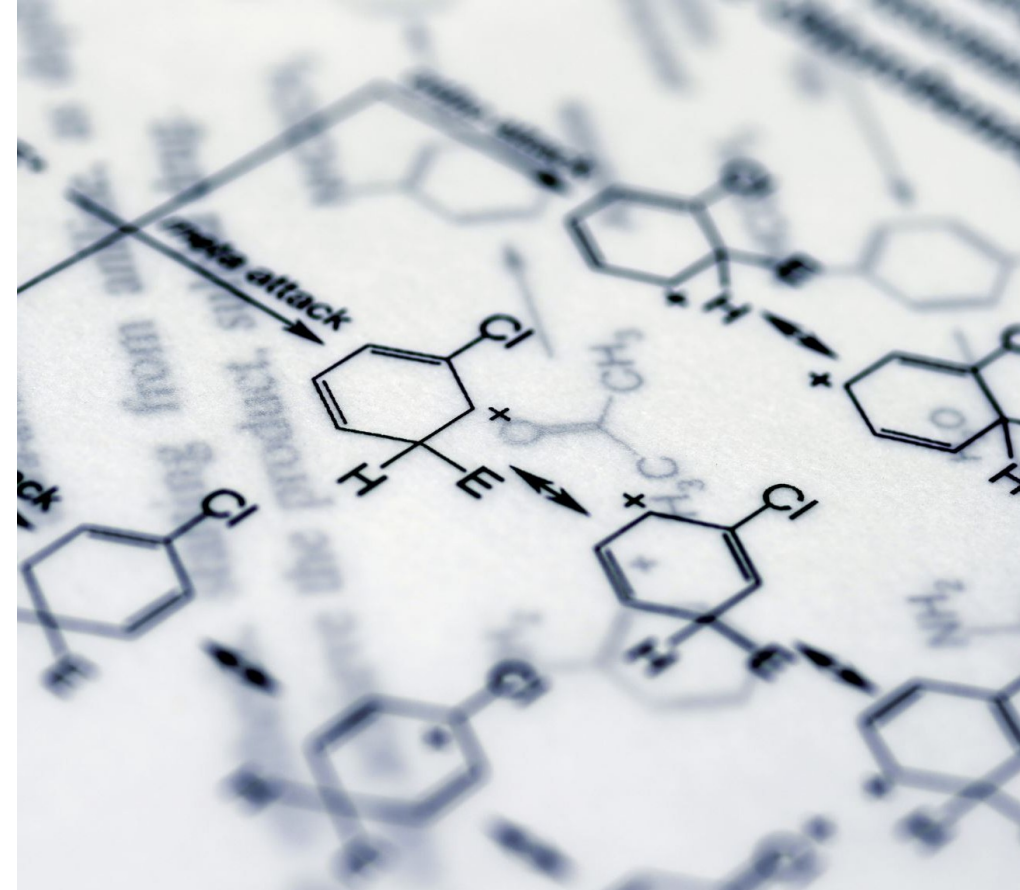
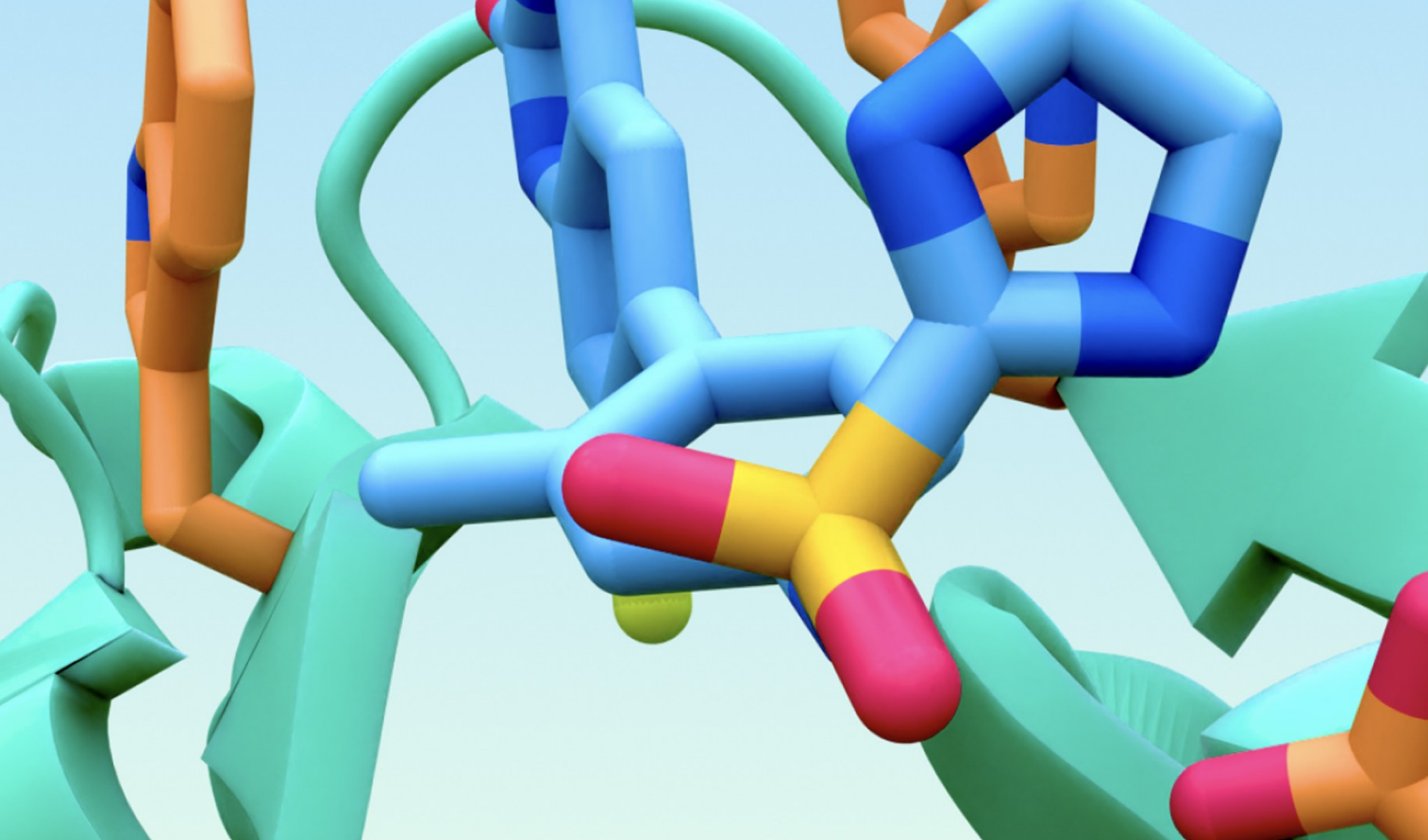




PROTEIN MODELLING AND STRUCTURE PREDICTION

Dr. Oluyemi Wande | Dr. Adeniyi Adewumi | Shadrach Eze | Stephen Nnemolisa



Sequence alignment

Blast

Filling Missing residues

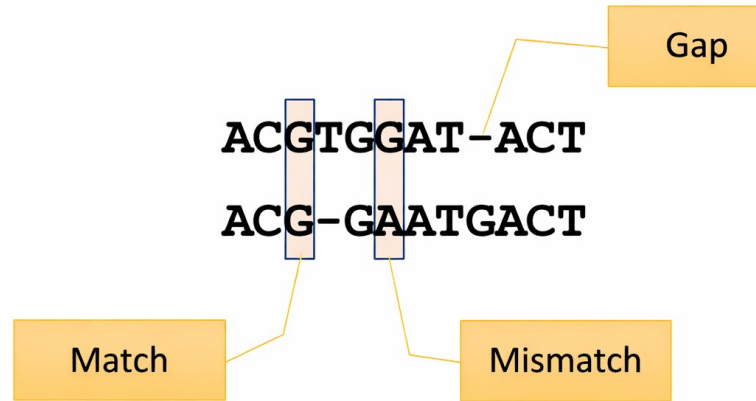
Swiss Model/Modeller

AlphaFold

Co-folding

Sequence Alignment

Alignment is the standard technique in molecular biology for comparing sequences



$$\text{Similarity}(\%) = \frac{\text{Match}}{\text{Match} + \text{Mismatch}} \times 100$$

$$\text{Above case: } \frac{9}{9 + 1} \times 100 = 90\%$$

Pairwise alignment

These similarities often indicate



- **Homology**: Two or more proteins share an ancestor.
- **Conserved Function**: The key amino acids or nucleotides that cannot or should not change.
- **Conserved Structure**: Regions that maintain a specific 3D shape.

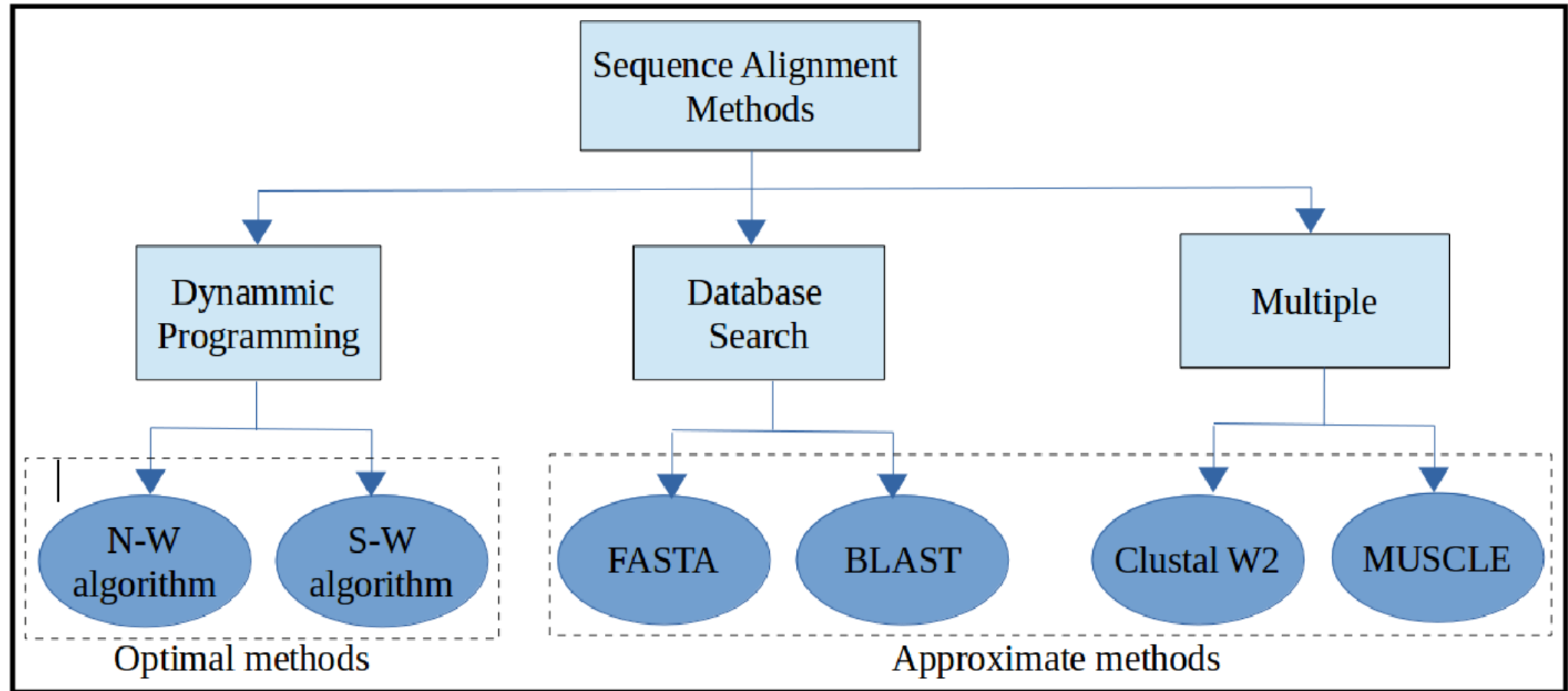
Multiple Sequence Alignment (MSA)

Aligning three or more protein/DNA sequences to understand conserved and non conserved regions

Histone H1 (residues 120-180)

HUMAN	KKASKPKKAASKAPT	KKPKATPVKKAKKK	LAATPKKAKKPK	TVKAKPVKASKPKKAKPVK
CHIMP	KKASKPKKAASKAPT	KKPKATPVKKAKKK	LAATPKKAKKPK	TVKAKPVKASKPKKAKPVK
MOUSE	KKAAPKKAASKAP	SKKPKATPVKKAKKK	PAATPKKAKKPK	VVKVPVKASKPKKAKTVK
RAT	KKAAPKKAASKAP	SKKPKATPVKKAKKK	PAATPKKAKKPK	IVKVPVKASKPKKAKPVK
COW	KKAAPKKAASKAP	SKKPKATPVKKAKKK	PAATPKKTKKPK	TVKAKPVKASKPKKTKPVK
	:	**:	*****:	*****:
NON-CONSERVED AMINO ACIDS	Conservative	Conservative	Non-conservative	Conservative
			Non-conservative	Semi-conservative
			Conservative	Non-conservative

Multiple Sequence Alignment



Kaur, Y., & Sohi, N. (2017). Comparison of Different Sequence Alignment Methods- A Survey. *International Journal of Advanced Research in Computer Science*, 8, 2308-2311.

Fasta files

```
>sp|P12345|EXAMPLE1_ECOLL Example protein 1 - E. coli  
MKKLLTLIIAGLVLALSVQAQETGTGSSGSSGSSGSSGGGGGGGGGGSSG  
SSGSGSSGSSGSSGSSGSSGSSGSSGSSGSSGSSGGGGGSSGSSGSSGSS  
GSSHHHHHHH
```

```
>tr|Q9BXR5|EXAMPLE2_HUMAN Example protein 2 - Human  
MATPSSLSFTSRAVLSRPRPRPRPRPRPRPRPRPRPRPRPRPRPTPRRAPTR  
PRPRPTPRPTPRPRPRPRPRPRPRPRPRPRPRPRPRPRPRPRPRPTPRRAPTRPR  
PRPRPRPRPRPRPRPRPRPRPRPRPRPRPRPRPRPRPRPRPRGRHHHHHH
```

```
>sp|Q8J2F4|EXAMPLE3_YEAST Example protein 3 - S. cerevisiae  
MSRVVFTGAGFGLGLALRNLLERGVTHVIVSRTESVAQDVVRQIQECGPRSVI  
LQADVTDEDVLRALAEESVTAVNGAGGNVRAVVSADRVADISVAVAGVGNSLI  
HARGGGSGGSGGGGSGGGVHHHHHH
```

BLAST

Enter your sequence

BLAST® » blastp suite

Standard Protein BLAST

blastn **blastp** blastx tblastn tblastx

BLASTP programs search protein databases using a protein query. more...

Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) [?](#) [Clear](#)

>sp|P12345|EXAMPLE1_ECOLL Example protein 1 - E. coli
MKKLLTLIIAGLVLALSVQAQETGTGSSGSSGSSGSSGGGGGGGGGGSSG
SSGSGSSGSSGSSGSSGSSGSSGSSGSSGSSGSSGGGGGSSGSSGSSGSS

From
To

Or, upload file No file chosen [?](#)

Job Title
Enter a descriptive title for your BLAST search [?](#)

☐ Align two or more sequences [?](#)

Choose Search Set

Database [?](#)

Organism
Optional
Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown. [?](#)

Program Selection

Algorithm ☒ blastp (protein-protein BLAST)
☐ PSI-BLAST (Position-Specific Iterated BLAST)
Choose a BLAST algorithm [?](#)

Search database **ClusteredNR** using **Blastp (protein-protein BLAST)**
☐ Show results in a new window

Select one or more organisms

Basic Local Alignment Search Tool

BLAST finds regions of similarity between biological sequences. The program compares nucleotide or protein sequences to sequence databases and calculates the statistical significance. [Learn more](#)

NEW

Mon, 21 Jul 2025
Here are a few highlights in our latest BLAST+ release:
Download BLAST+ 2.17.0 now! [More BLAST news...](#)

Web BLAST

Nucleotide BLAST
nucleotide ► nucleotide

blastx
translated nucleotide ► protein

tblastn
protein ► translated nucleotide

Protein BLAST
protein ► protein

ResLaR
Research Laboratory for Rational Design
of Drugs and Biomaterial

6

SEQUENCE ALIGNMENT

#Update the package list and install clustalo

```
cat > test.fasta << 'EOF'
sudo apt-get update
sudo apt-get install clustalo
```

#check the version

```
>seq2
>seq3
clustalo --version
EOF
```

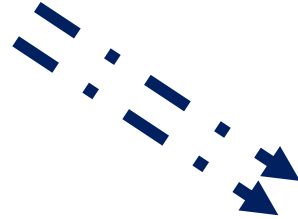
#create a sample multisequence file

```
cat > test.fasta << 'EOF'
>seq1
MKKLLTLIIAGLVLAL
>seq2
MKKLLTLIIAGLALAL
>seq3
MKKLLTLIIAGLVLSL
EOF
```

#run the clustering

```
clustalo -i test.fasta -o test.aln -v
```

- **ClustalW** is the original version released in 1994
- Has been superseded by the newer **ClustalOmega** release in 2011
- **ClustalOmega** is faster for even sequences up to 1000+ length



CLUSTAL 2.1 multiple sequence alignment

```
seq1
seq3
seq2
MKKLLTLIIAGLVLAL
MKKLLTLIIAGLVLSL
MKKLLTLIIAGLALAL
*****.*:*
```



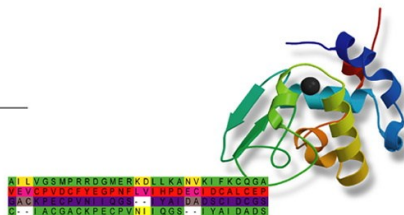
PDB_FIXER



FILLING MISSING RESIDUES

Modeller

Program for Comparative Protein
Structure Modelling by Satisfaction
of Spatial Restraints



PRIME

Filling Missing Residues



PDB_FIXER

```
#Update the package list and install clustalo
```

```
sudo apt-get update  
conda install -c conda-forge pdbfixer
```

```
#check the version
```

```
pdbfixer --version
```

```
#run the fixing on a sample pdb structure
```

```
pdbfixer 9NSR.pdb --add-residues --add-atoms --add-hydrogens --ph=7.0
```

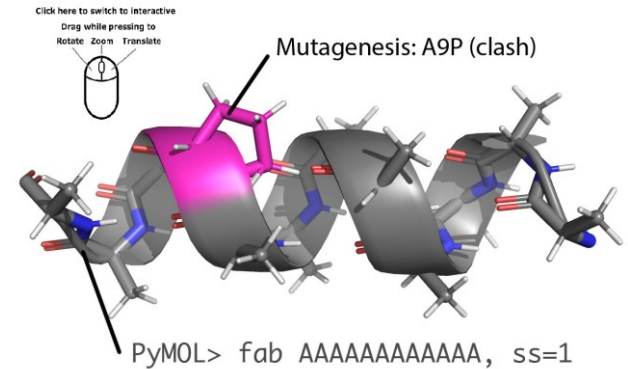
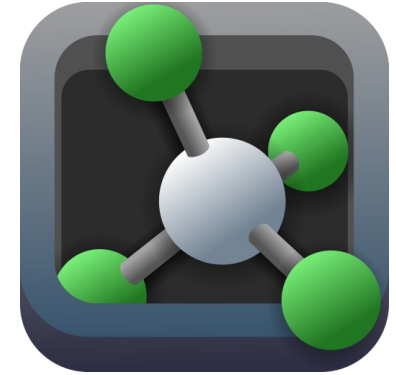
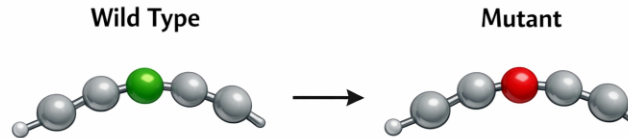
```
#use help to see more options
```

```
pdbfixer --help
```

```
usage: pdbfixer [-h] [--pdbid PDBID] [--url URL] [--output OUTPUT]  
               [--add-residues] [--add-atoms {all,heavy,hydrogen,none}]  
               [--replace-nonstandard] [--ph PH]  
               [--water-box X Y Z] [--positive-ion {Na+,K+,Cs+,Li+,Rb+}]  
               [--negative-ion {Cl-,Br-,F-,I-}] [--ionic-strength STRENGTH]  
               [--verbose] [--version]  
               [filename]
```

**Can also be
used via a
python API**

Mutating your Protein



swapaa :248 thr

- **AlphaFold (with mutated sequence)**
- **ESMFold (with mutated sequence)**

Comparative/Homology Modelling



Predicts the 3D structure of a protein based on a similar proteins with known experimental structure



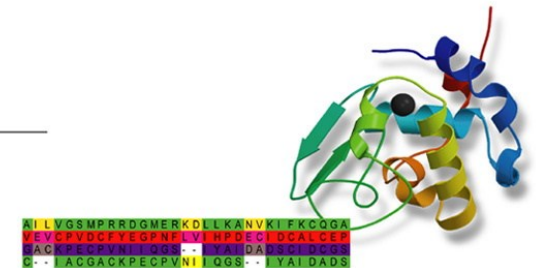
If two proteins share similar sequences, they likely share similar 3D structures



SWISS-MODEL

Modeller

Program for Comparative Protein
Structure Modelling by Satisfaction
of Spatial Restraints



Start a New Modelling Project

Target Sequence(s):
(Format must be FASTA, Clustal,
plain string, or a valid UniProtKB AC)

Paste your target sequence(s) or UniProtKB AC here

+ Upload Target Sequence File...

Validate

Project Title:

Untitled Project

Email:

Optional

Search For Templates

Build Model

Supported Inputs

Sequence(s)

Target-Template Alignment

User Template

DeepView Project

By using the SWISS-MODEL server, you agree to comply with the following [terms of use](#) and to cite the corresponding [articles](#).

**Get the Target
sequence**

**Search for
template with
high similarity**

**Sequence
alignment**

**Quality Control
Post-processing**

Modeller

Program for Comparative Protein
Structure Modelling by Satisfaction
of Spatial Restraints

A I L V G S M P R R D G M E R K D L L K A N V K I F K C Q G A
V E V C P V D C F Y E G P N F L V I H P D E C I D C A L C E P
A C K P E C P V N I L Q G S . . Y A I D A D S I D C G S
G . . A D G A C K P E C P V N I L Q G S . . Y A I D A D S



Install via command
line or download the
executable for your
operating system

Works on
command line
only --- **python**
API only



Requires three files:

- Alignment file
- Template structure
- Python script

Reports the **Molecular
Probability Density
Function (molpdf)** as an
objective function

Can be messy but
have a workaround in
UCSF Chimera



Python script

```
## Comparative modeling by the AutoModel class
from modeller import *
from modeller.automodel import * # Load the AutoModel class

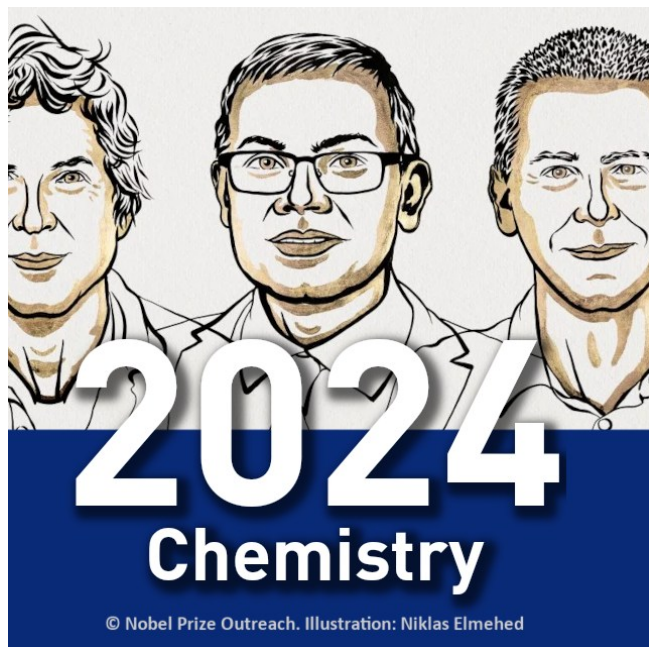
log.verbose()
env = Environ()

# directories for input atom files
env.io.atom_files_directory = ['.']

a = LoopModel(env,
               alnfile = 'alignment.ali', # alignment filename
               knowns   = '9nsr_1',      # codes of the templates
               sequence = 'chain_A')     # code of the target
a.starting_model = 1 # index of the first model
a.ending_model   = 5 # index of the last model
                  # (determines how many models to calculate)
a.md_level = None  # No refinement of model

a.loop.starting_model = 1 # First loop model
a.loop.ending_model   = 4 # Last loop model
a.loop.md_level        = refine.fast # Loop model refinement level

a.make() # do comparative modeling
```



THE REVOLUTION: ALPHAFOLD

- Can now predict virtually any protein structure by just using the primary sequence
- Just a few limitations with post-translational modifications and cannot model dynamics

Remaining jobs: 30

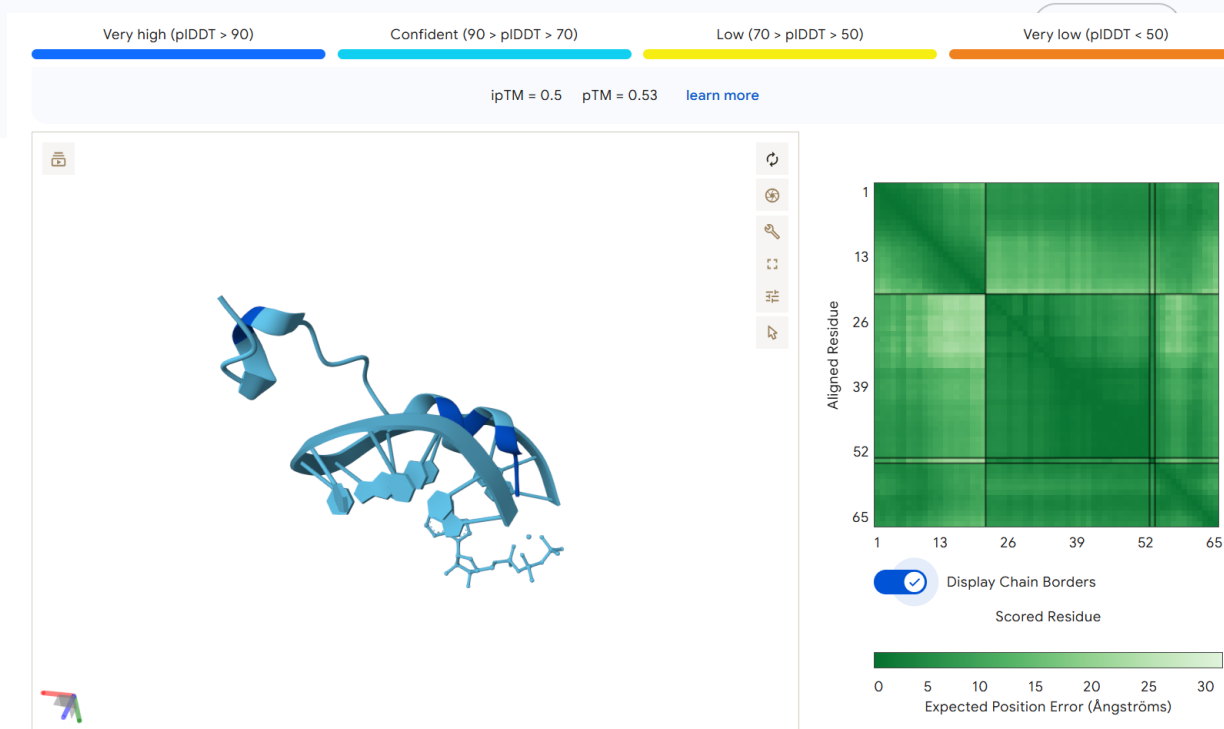
AlphaFold Server allows you to model a structure consisting of many biological molecules

[Learn more](#)

[Upload JSON](#) [Clear](#)

Entity type Protein	Copies 1	Q S P D I A Q G V H L D R N E E D I G A G	⋮	⌵
Entity type Ligand	Copies 1	ATP – Adenosine triphosphate	⋮	⌵
Entity type Ion	Copies 1	Mg ²⁺	⋮	⌵
Entity type DNA	Copies 1	>Paste sequence or fasta ATTCTCATTCAQ	⋮	⌵

+ Add entity



Uncertainty measures

pLDDT

pLDDT (predicted Local Distance Difference Test score)

- Measures certainty about the position of each residue

ipTM

ipTM (interface predicted Template Modelling score)

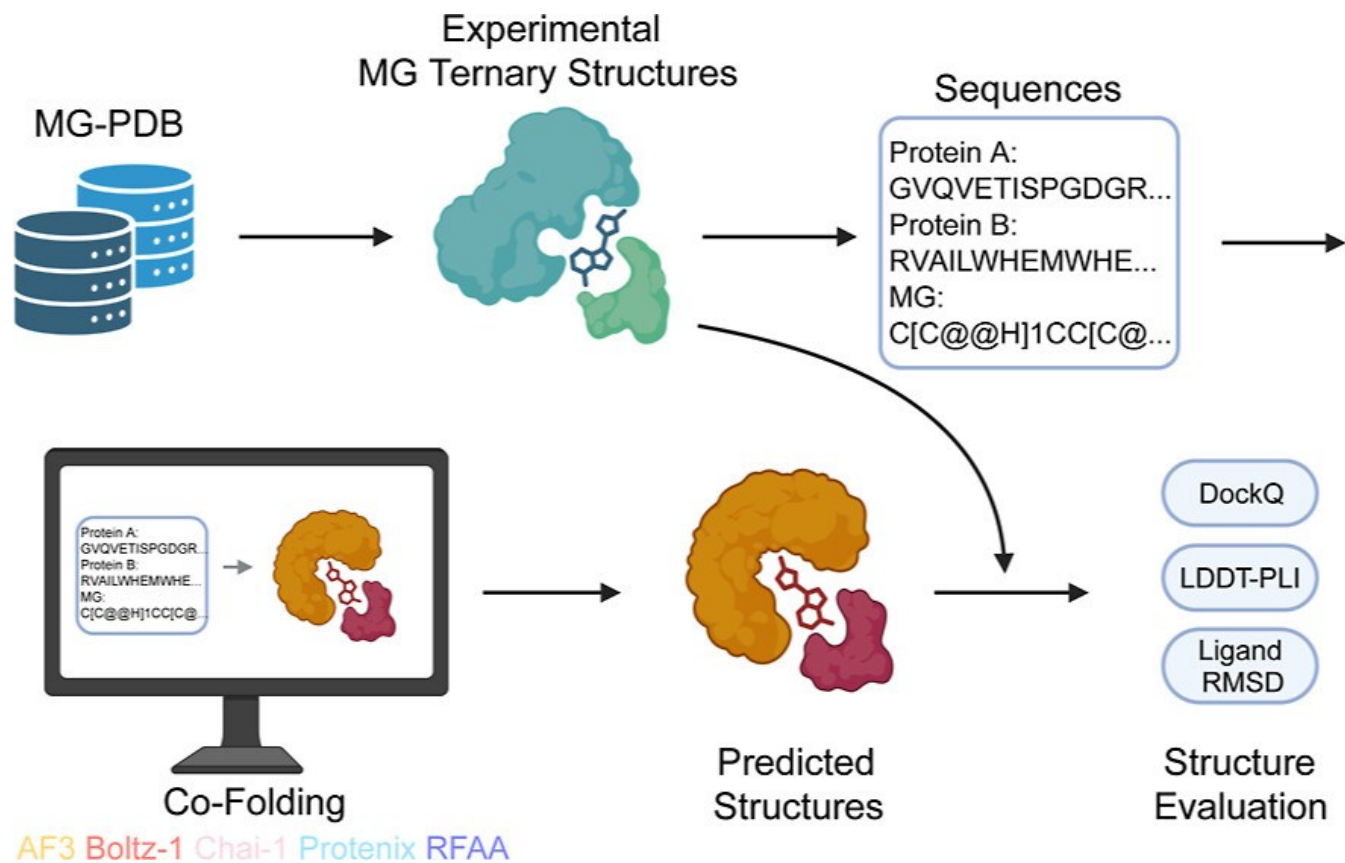
- How accurately one or more chains interact with each other

pTM

pTM (predicted Template Modelling score)

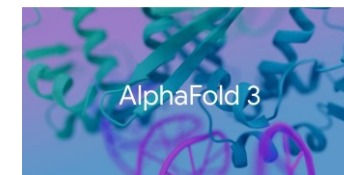
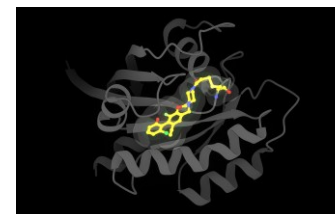
- How accurate the global prediction of each chain is

Co-Folding



<https://pubs.acs.org/doi/10.1021/acs.jcim.5c01860>

What if we can bypass docking and predict protein structure and ligand binding at same step?



Further References

- <https://home.cc.umanitoba.ca/~psgendb/tutorials/bioLegato/multalign/multalign.html>
- <https://htmlpreview.github.io/?https://github.com/openmm/pdbfixer/blob/master/Manual.html>
- <https://swissmodel.expasy.org/>
- Webb, B., & Sali, A. (2016). Comparative Protein Structure Modeling Using MODELLER. *Current protocols in bioinformatics*, 54, 5.6.1–5.6.37. <https://doi.org/10.1002/cpbi.3>
- <https://alphafoldserver.com/welcome>
- Jumper, J., Evans, R., Pritzel, A. *et al.* Highly accurate protein structure prediction with AlphaFold. *Nature* **596**, 583–589 (2021). <https://doi.org/10.1038/s41586-021-03819-2>